

Anirudh Prasad

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Summary

Security researcher and systems engineer focused on vulnerability research across Linux kernels, blockchain virtual machines, smart contracts, and low-level systems software. Experienced in C, C++, Rust, fuzzing, reverse engineering, kernel debugging, and distributed systems. Discovered and upstreamed **CVE-2026-89587**, a Linux kernel ACPI stack buffer overflow, and have performed security research against production blockchain infrastructure and widely deployed software.

Education

PES University

B.Tech. in Computer Science and Engineering

Bengaluru, India

2022 – 2026

Experience

Movement

Remote

Systems Engineer III

Dec 2024 – Sept 2026

- Worked on Aptos-derived blockchain infrastructure spanning the Move VM, consensus, validator execution, networking, protocol economics, and core node systems, primarily in Rust and Move.
- Built fuzzing infrastructure for the Move VM including crash replay and triage, semantic fuzzing, and economic invariant checking; used the infrastructure to identify and remediate a global buffer overflow in Move value deserialization.
- Hardened VM type construction against type-depth and type-size bypasses in `PackClosure`, adding validation around `TypeBuilder::create_function_ty`.
- Hardened consensus and networking paths against resource-exhaustion and denial-of-service conditions through DKG blob validation, proposal metadata limits, per-peer message/byte throttling, transaction rate limiting, and a patch for **CVE-2024-32650** in `rustls`.
- Improved validator reliability by replacing panic-prone state-transition paths with recoverable errors, fixing a state-sync fork race, and preventing database truncation leaks; added tests scaling state transitions to 500 accounts.
- Implemented a staking-rewards failsafe in Move to prevent reward distributions from exceeding available protocol balances, with unit and end-to-end validation.
- Developed production observability infrastructure using Prometheus, Grafana, OpenTelemetry, and Rust tracing across full-node and consensus components.

Nokia

Remote

Software Engineering Intern – NCMT

Aug 2025 - Nov 2025

- Worked on Nokia Core Monitoring and Troubleshooting (NCMT), developing Go-based monitoring and alerting components using Prometheus, Grafana, Kafka, and Python.
- Independently developed a Python-based ML-related ingestion engine for telecom operational data.

Eclipse Labs

Remote

Developer

Dec 2023 – Apr 2024

- Developed Rust smart contracts and TypeScript/JavaScript components for the Eclipse blockchain testnet and mainnet, working across Solana-based infrastructure and smart-contract tooling.
- Participated in code reviews and engineering design while maintaining technical documentation for blockchain infrastructure and developer workflows.

Security Research

Linux Kernel – CVE-2026-89587

Upstreamed

ACPI Platform Firmware Runtime Update

2026

- Discovered and upstreamed a stack-based buffer overflow in the Linux kernel's ACPI Platform Firmware Runtime Update driver, caused by unchecked firmware-supplied ACPI `_DSM` buffer lengths in `query_capability()`.
- Reproduced three KASAN stack-out-of-bounds reports on Linux 7.2-rc6 using oversized DSM buffers; fixed the issue through per-field bounds validation before copying firmware-controlled buffers.

.NET System.Formats.Tar – Symlink Escape

Independent Research

Path Traversal / Extraction Security

2026

- Discovered a symlink-target bounds-check bypass in `TarFile.ExtractToDirectory` on .NET 9.0.318, bypassing protections introduced for CVE-2026-45491 through archive-created intermediate symlinks.
- Developed a five-entry malicious PAX tar archive that exploits the discrepancy between lexical and physical symlink resolution, causing a symlink under the extraction directory to resolve to `/etc`; verified access to `/etc/passwd` through the resulting symlink chain.

Move VM Security Research

Fuzzing, Memory Safety, and Type-System Security

Movement Labs

2025 – Present

- Built semantic and invariant-driven fuzzing infrastructure for the Move VM, including crash reproduction, triage, and economic invariant validation.
- Investigated memory-safety and type-system attack surfaces in the Move execution environment, including deserialization and function-type construction paths.

Independent Vulnerability Research

Web Security and Responsible Disclosure

2023 – Present

- Reported security issues affecting public web infrastructure, including an HTML injection issue on a Stanford University subdomain and a JIRA configuration issue affecting Cambridge University infrastructure.
- Reported a security issue through NASA's vulnerability disclosure program and received official acknowledgement from NASA.
- Conducted a full penetration test against PES University's grading portal and student-faculty portal, successfully uncovering many critical vulnerabilities. Provided fixes and mitigations with zero downtime or disruption.
- Conducted ongoing kernel and systems vulnerability research using Ghidra, WinDbg, Driver Verifier, GDB/LLDB, fuzzing, dynamic analysis, and source-level debugging.

Selected Projects

IcarOS

github.com/0xIcarus/IcarOS

x86 Assembly, C, Bash

- Built an x86 operating-system project from a custom bootloader and low-level C/assembly components, developing practical understanding of machine-level execution and OS internals.

Yocto Compiler

github.com/0xIcarus/Yocto

OCaml, LLVM

- Implemented a small compiler using OCaml and LLVM, exploring lexical analysis, parsing, intermediate representations, and code generation.

Merkle Trees for ZK Circuits

github.com/0xIcarus/merkle-integration

Rust

- Implemented Merkle-tree primitives in Rust for integration with zero-knowledge proving circuits.

Socketeer

github.com/0xIcarus/Socketeer

Python, Linux

- Developed a Linux network-security framework implementing ARP/DNS spoofing and HTTP traffic redirection for controlled security research.

Recognition & Community

- Placed 3rd nationally in a CTF competition; participated in competitive security research and capture-the-flag competitions.
- Delivered a Web Security and Reverse Engineering lecture at IIIT Bangalore.
- Teaching Assistant at PES University, mentoring students in security, systems, and software engineering.

Technical Skills

Languages: C, C++, Rust, Go, Python, Bash, OCaml, JavaScript/TypeScript, Solidity, x86 Assembly

Security Research: Vulnerability Research, Fuzzing, Memory Safety, Kernel Security, Binary Analysis, Reverse Engineering, Exploit Analysis, Web Security, Smart-Contract Security, Responsible Disclosure

Systems: Linux Internals, Kernel Debugging, Operating Systems, Virtual Machines, Compilers, LLVM, Networking, Concurrency, Memory Management, Distributed Systems

Blockchain: Move, Rust Smart Contracts, Solana, Consensus, Merkle Trees, Zero-Knowledge Proving Concepts, Distributed Systems

Tools: GDB, LLDB, WinDbg, Driver Verifier, Ghidra, radare2, Wireshark, strace, perf, Git, Docker, Prometheus, Grafana, OpenTelemetry